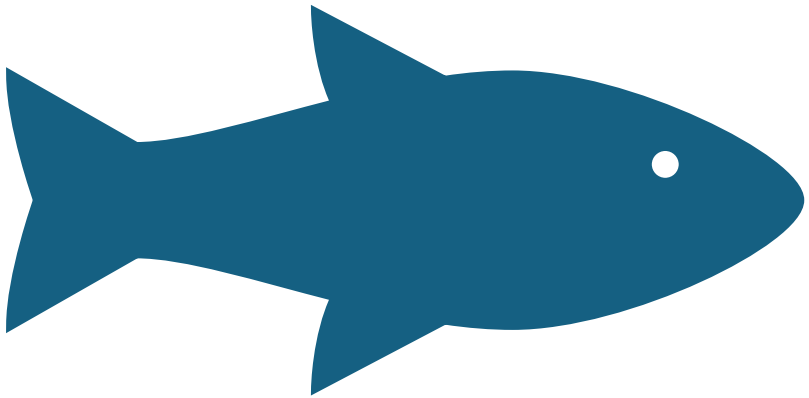


1.6 million km²

The estimated size of the Great Pacific Garbage Patch

2.41 million tonnes/year

The amount of plastic entering the oceans from rivers every year



Marine Microplastic Accumulation

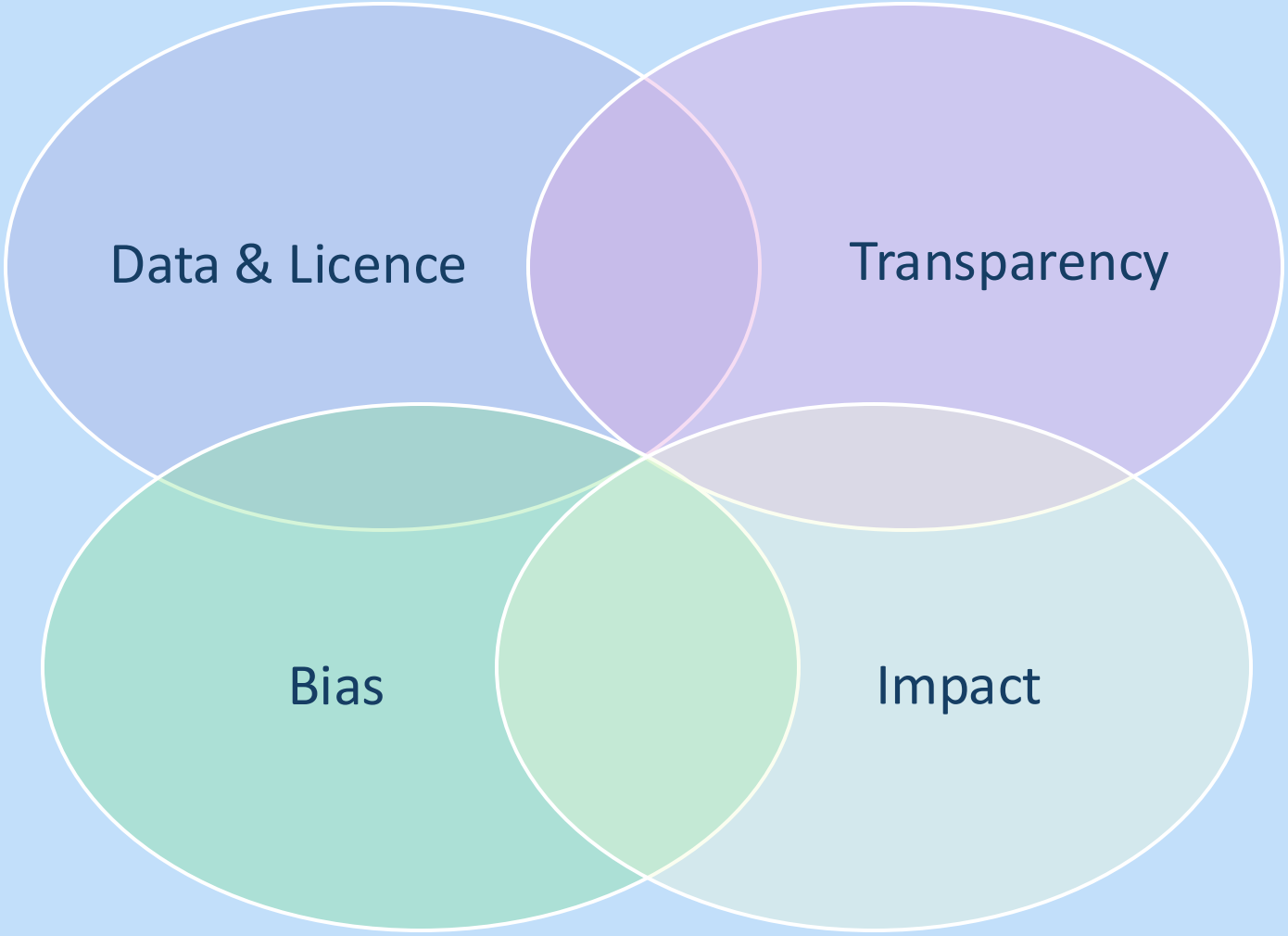
Can ocean currents predict future marine microplastic accumulation hotspots?

20 425 marine microplastic sample

Source : marine microplastic database, via kaggle

Where ?	When & How ?	How much ?
• Latitude • Longitude • Oceans • Regions • Subregions • X and Y	• Date • Sampling method • Organization • References, DOI	• Measurement • Unit • Density class

Ethical concerns



Ethical concerns

Data & Licence

- Public data, via Kaggle
- Source cited

Transparency

- Different sampling methods across studies

Bias

- Mostly North Atlantic
- Uneven over time
- Different tools (nets, buckets) capture different things

Impact

- An empty area on the map is not a clean area
- A hotspot may reflect where people looked

Missing values in the Dataset

Missing measurements: 5812 values missing

Missing regions: 11540 values missing

Missing subregions: 19118 values missing

Missing oceans: 271 values missing

Data cleaning

Duplicates; None found

Unnecessary columns; Removed references, IDs & duplicate coordinate details

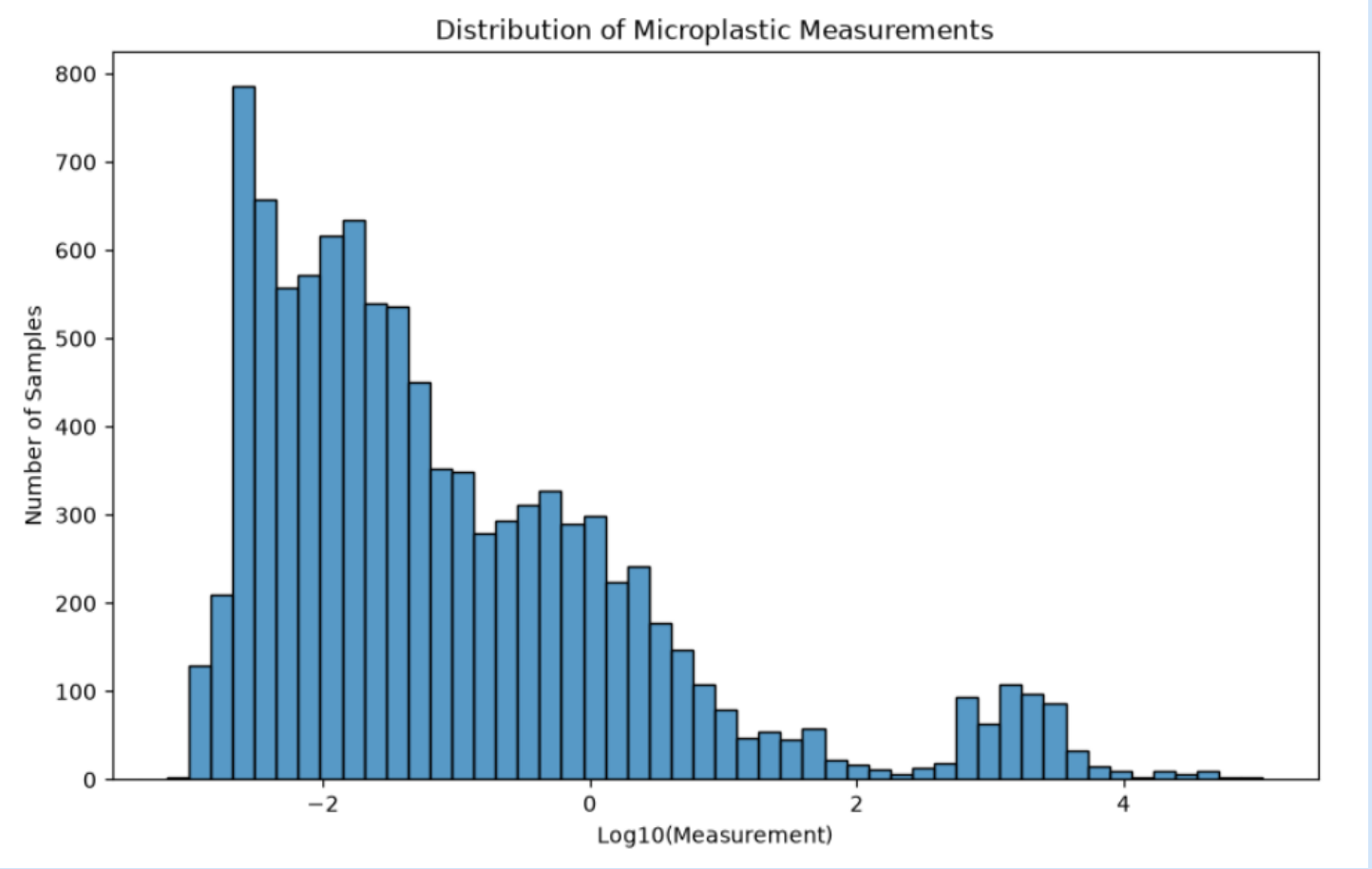
Missing values; Filled missing ocean and region values using nearby sampling locations

Missing measurements; Left unchanged rather than replaced with 0

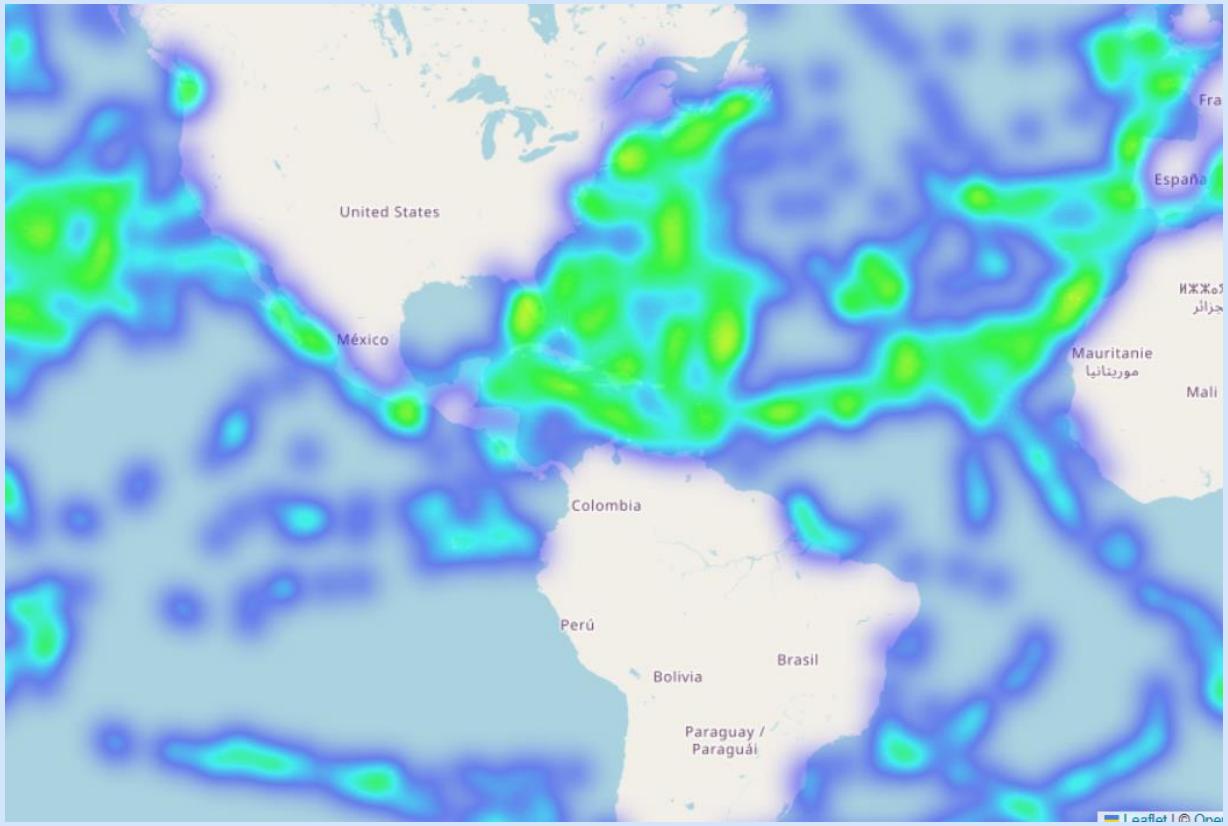
Data quality; Reviewed summary statistics to identify unusual measurements

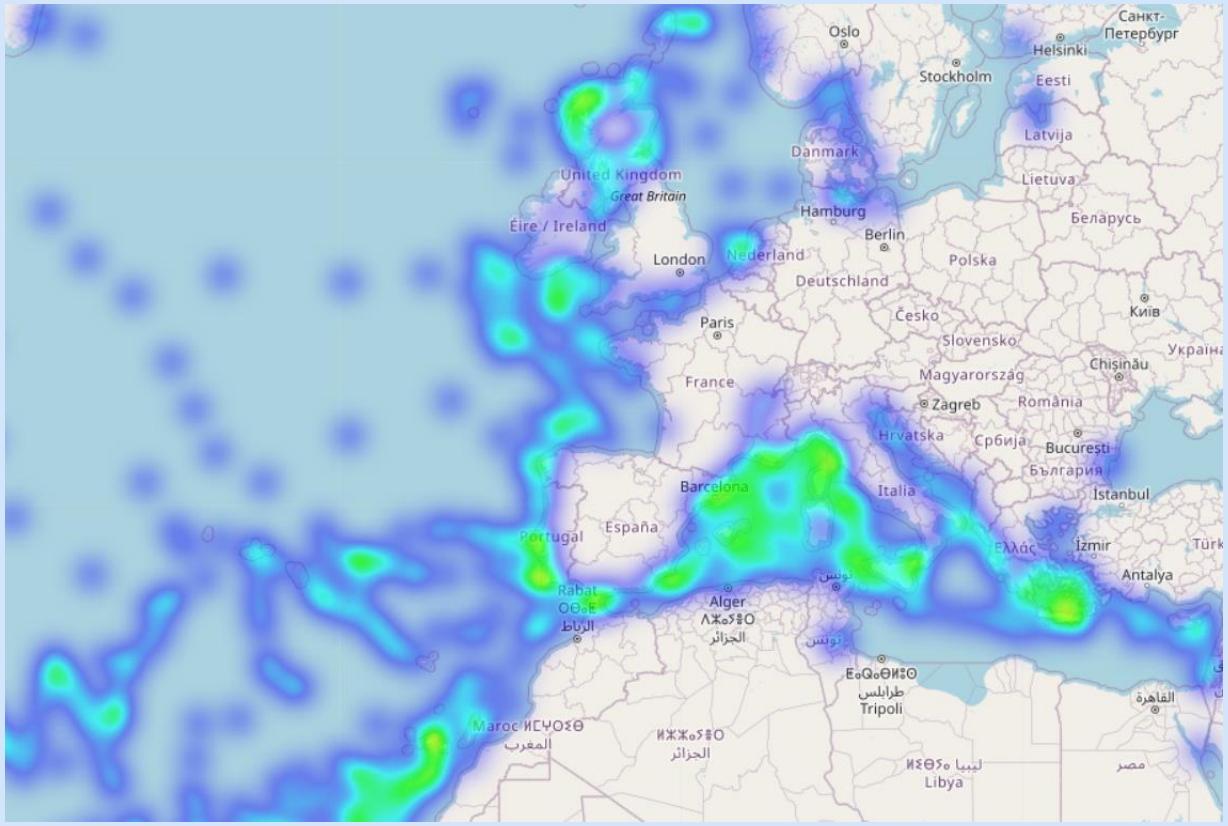
Concentration distribution

- Most measurements (microplastic conc. per cubic metre) relatively low
- Right-skewed distribution: very few high values
- Small cluster of measurements around 10^3

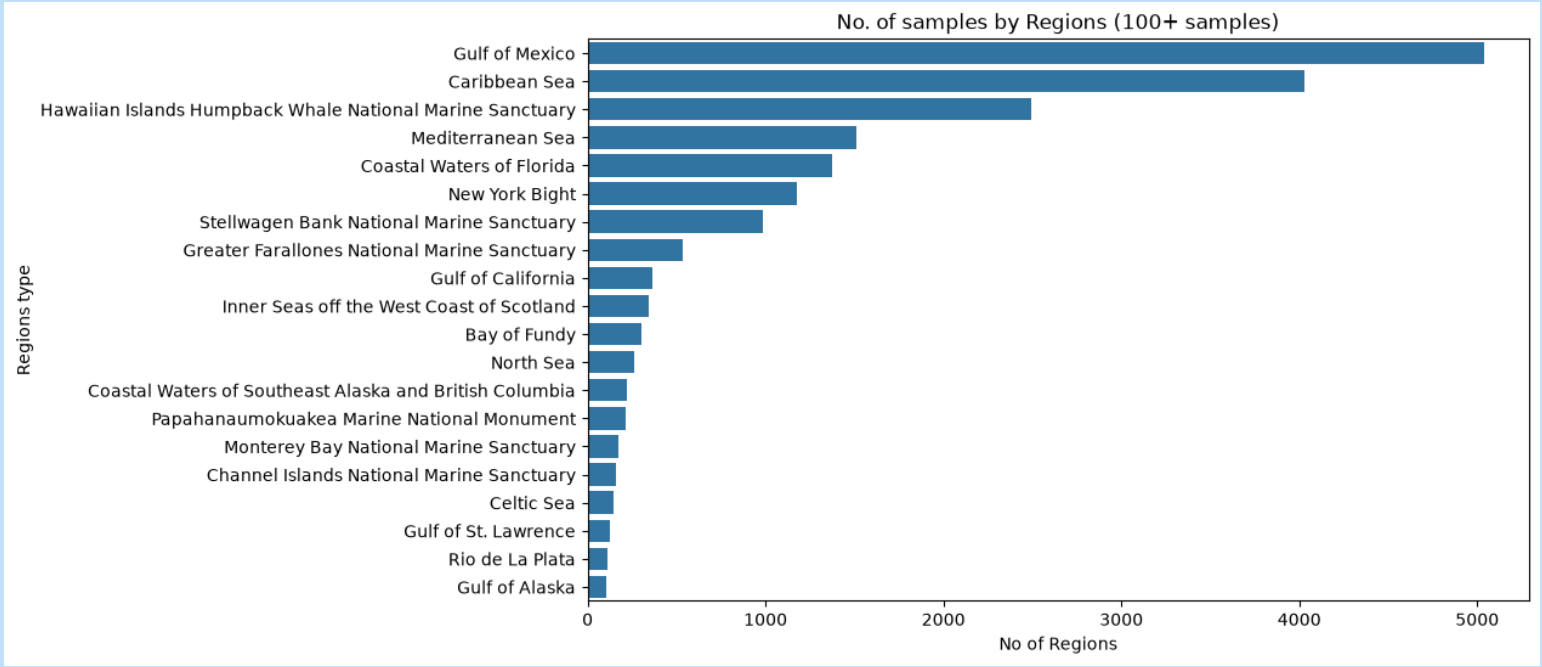


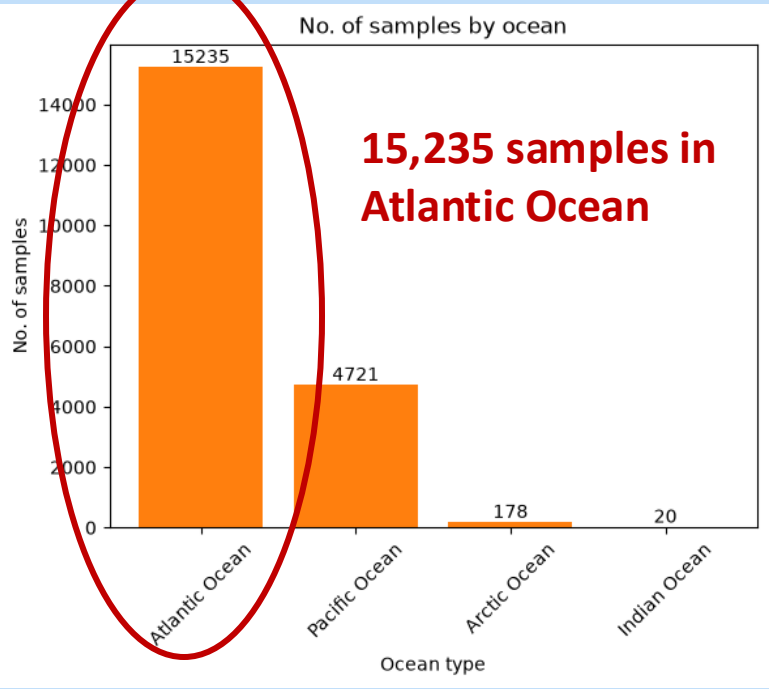
Global Distribution of Marine Microplastic Hotspots





Descriptive Statistics






```
Measurement column statistics:
count    14613.00
mean       161.98
std       2198.86
min          0.00
25%          0.00
50%          0.01
75%          0.13
max     110480.00
Name: Measurement, dtype: float64
```

Wide range of sample measurements

Larger number of samples (20,425), however, large chunks of data missing

Limits and next steps

Limits

Biased Coverage

Samples are concentrated in the Atlantic and near costs

Traning set : Atlantic only

Where our data is densest

Next steps

- **Add ocean currents**
- **Match samples to currents**

Sources

- Image diapo 1 : déchets plastiques: Nos derniers articles et actualités - L'Humanité : lire, agir. (n.d.). *L'Humanité*. Retrieved 11 October 2026, from https://www.humanite.fr/mot-cle/dechets-plastiques
- Marine Microplastic Dataset: *Kaggle: The World's AI Proving Ground*. (n.d.). Retrieved 11 October 2026, from https://www.kaggle.com/